

SASIKIRAN SIVAKUMAR

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Software Developer | Java, Spring Boot, React, Cloud Technologies | AWS Certified

EDUCATION

Master of Applied Computer Science

Concordia University

September 2023 – June 2025

Montreal, QC

Coursework: Applied Artificial Intelligence, Machine Learning, Data Structures, Algorithm Design Techniques, Distributed Systems Design

B.Tech Computer Science (AI/ML)

SRM Institute of Science and Technology

June 2019 – May 2023

Chennai, India

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, TypeScript, HTML5, CSS

Frameworks & Libraries: Spring Boot, ReactJS, Express.js, Maven, Angular

Cloud & DevOps: AWS (EC2, S3, Lambda, CloudWatch), Azure, Docker, Kubernetes, Jenkins, Git

Databases & Data Technologies: PostgreSQL, MongoDB, NoSQL, SQL

Development Practices: Object-Oriented Design, Design Patterns, RESTful APIs, Microservices, Unit Testing

Languages: Bilingual fluency (**French - (DELF B2) & English**)

EXPERIENCE

Research Intern | Publication

MIT Square | Tensorflow, Python

July 2022 – October 2022

Chennai, India

- Conducted advanced research on LSTM neural network architectures for weather forecasting using Python and **TensorFlow**, processing 50,000+ historical time-series data points
- Designed and implemented a **Convolutional LSTM** (ConvLSTM) neural network model featuring 130,811 trainable parameters, utilizing ReLU activation functions and Adamax optimizer for enhanced prediction accuracy
- Optimized model performance through systematic hyperparameter tuning, feature engineering increasing predictive accuracy by 10% and reducing mean absolute error by 15%

PROJECTS

Event-Driven Task Tracker | Java, Spring Boot, Apache Kafka, Docker, PostgreSQL

GitHub

- Architected and developed a Microservices-based Task Management system using Java and **Spring Boot** with **PostgreSQL** as database, implementing Event-Driven architecture for real-time task processing
- Implemented Robust API Security and Messaging by integrating Spring Security with **JWT** auth while orchestrating real-time, asynchronous event communication between services via **Apache Kafka**
- Automated infrastructure deployment using Docker containerization and Docker Compose, reducing local development setup time by 80%

Risk Game | Java, Design Patterns, Object-Oriented Programming, Maven, JUnit4

GitHub

- Engineered a sophisticated turn-based strategy game engine applying 4 GoF **Design Patterns** to manage complex game phases
- Collaborated within a 6-member Agile team, maintaining code quality and integrity through Git version control, automated CI/CD pipelines with **GitHub Actions**, and comprehensive **JUnit4** unit testing achieving 90% code coverage
- Architected system using core **Object-Oriented Programming** principles and implementing Observer Pattern for decoupled event logging with complete Javadoc documentation

InventoryEase | JavaScript, React, Node.js, Express.js, MongoDB, HTTP

GitHub

- Developed a full-stack **MERN** product inventory management system serving 1,000+ simulated products
- Developed and tested a **REST API** supporting CRUD operations and database interactions, ensuring reliability and fault tolerance
- Engineered a responsive **ReactJS** frontend with component-based architecture, implementing real-time state management, improving user experience and reducing page load time by 40%

CERTIFICATIONS

Certified Cloud Practitioner

Amazon Web Services

Issued May 2025